



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

MS4672 - Week 6

Bayesian Optimization for Materials Discovery

From Gaussian processes to experiments with the OT-2

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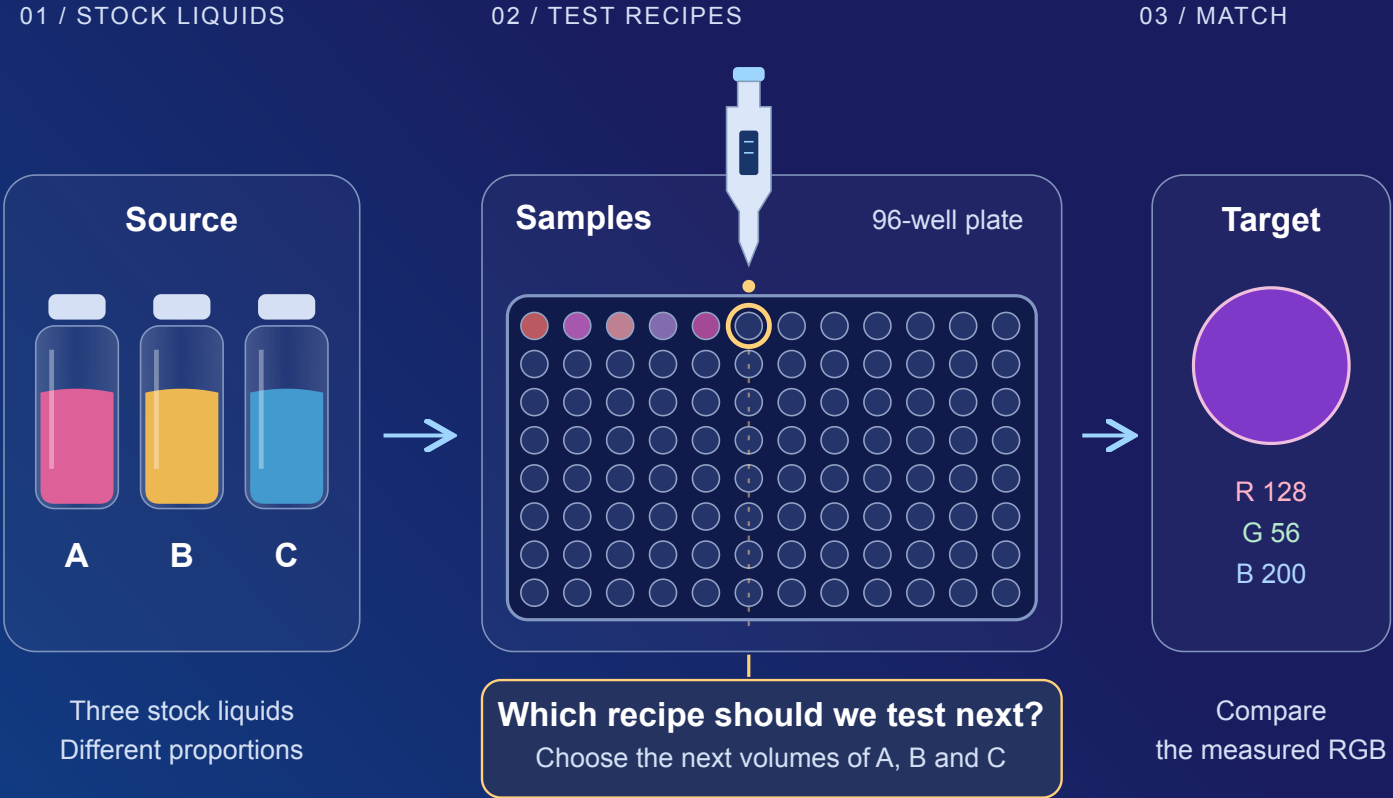
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One target. Many possible recipes.

Mix A, B and C to match a target color.
Every test costs liquid, time and one well.

Bayesian optimization learns from the tested recipes to choose the next experiment.

Mix → measure → compare



Illustrative colors and well positions. The notebook measures its target from a reference well.

The recipe is *our input x* .

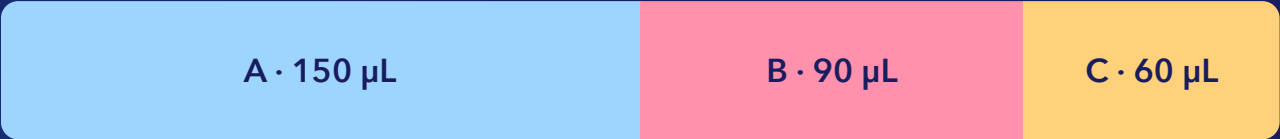
The robot needs volumes. The optimizer uses fractions. Each recipe must satisfy the fixed total.

$$\underline{V}_A + \underline{V}_B + \underline{V}_C = \underline{V}_{\text{tot}} = 300\mu\text{L}$$
$$\underline{x} = \frac{1}{\underline{V}_{\text{tot}}} [\underline{V}_A, \underline{V}_B, \underline{V}_C]^T$$

Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.

Only two fractions are independent; the third is fixed by the total.

One recipe, two representations



Liquid	Volume	Fraction
A	150 μL	0.50
B	90 μL	0.30
C	60 μL	0.20

Illustrative recipe. Stock-liquid identities A, B and C are independent of RGB channels.

MSE is our *optimization score*.

Square the difference from the target in each RGB channel, then take the average. The optimizer learns from and minimizes this MSE.

$$\text{MSE} = \frac{(\Delta R)^2 + (\Delta G)^2 + (\Delta B)^2}{3}$$

Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.

Y stores MSE: a smaller score means a closer RGB match.

A worked RGB example

Channel	Target	Sample	Difference ²
R	128	134	6 ² = 36
G	56	50	(−6) ² = 36
B	200	206	6 ² = 36

OPTIMIZATION SCORE · MSE

36

(36 + 36 + 36) / 3 = 36

Visualization note: RMSE = √MSE = 6. RMSE will be used for visualization because its unit is the same as the RGB channel values.

Illustrative data: target [128, 56, 200], sample [134, 50, 206].

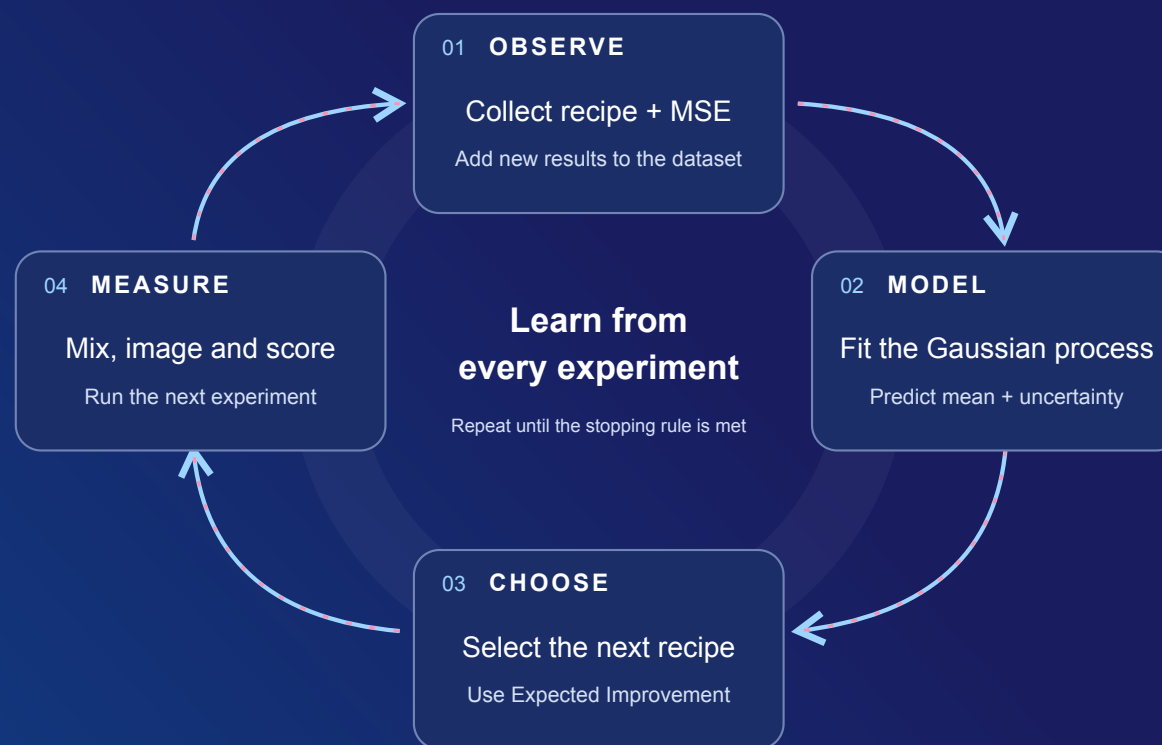
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The Bayesian *optimization cycle.*

Use a small initial dataset to build a model. Choose a new recipe, measure it, and update the model.

The new measurement returns to the dataset, then the GP is fitted again.

A closed loop: every result informs the next choice



Each measured result becomes evidence for the next decision.

Follow the arrows clockwise. Select any stage to explain its role.

GP 01 · BEFORE THE DATA

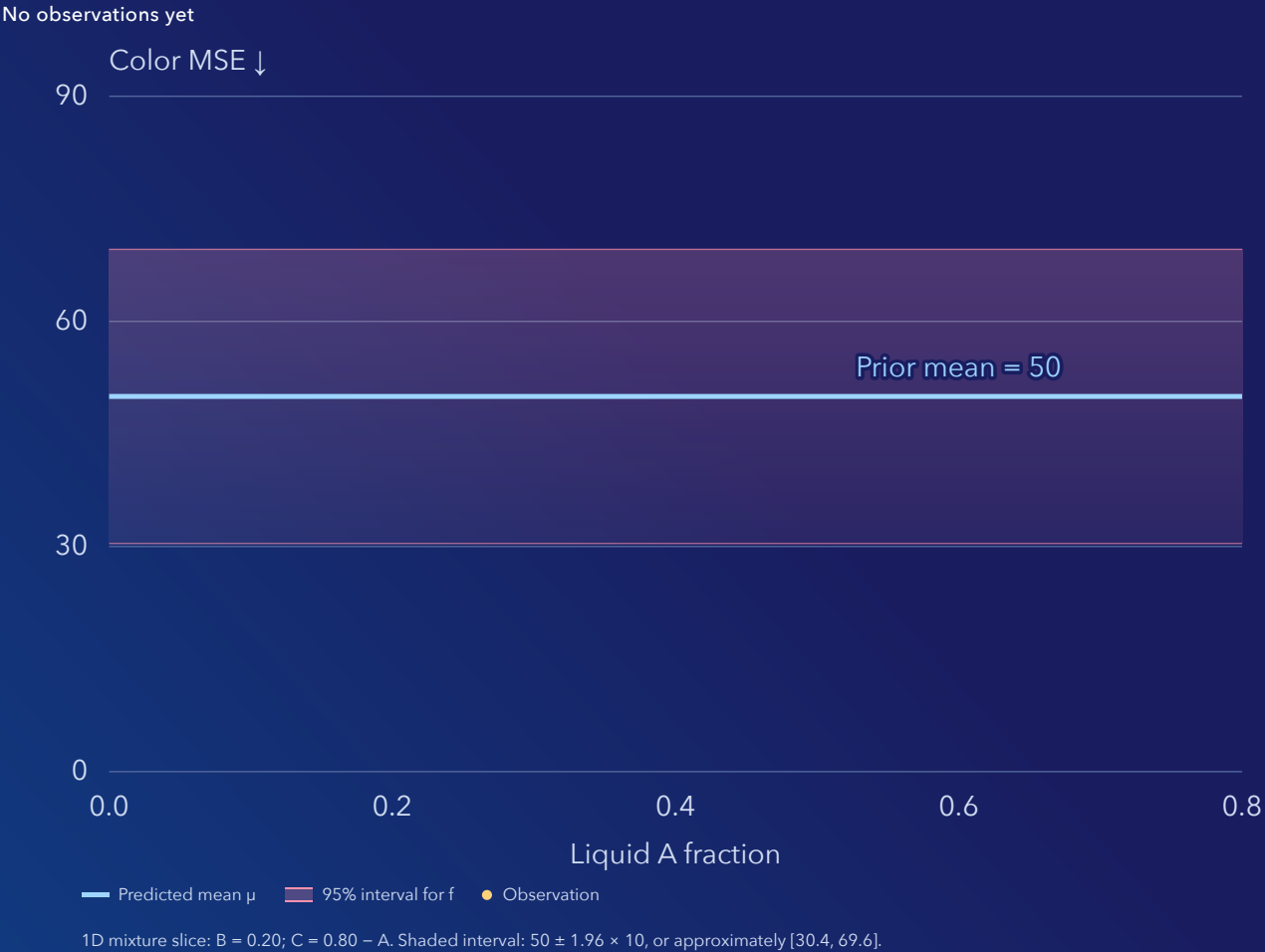
Start with a *prior belief*.

Before measuring anything, suppose the underlying MSE is **around 50**, with standard deviation 10. Every recipe starts equally uncertain.

$$\underline{m}(\underline{x}) = 50$$
$$\underline{k}(\underline{x}, \underline{x}) = 100 = 10^2$$

Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.

A prior is a starting assumption. The data will update it.



GP 02 · ONE OBSERVATION

One measurement *bends the prediction.*

We test $A = 0.20$ and observe $\text{MSE} = 30$. The model moves toward that observation and becomes more certain nearby.

$$y_{\cdot 1} = 30$$

$$y_i = f(x_i) + \varepsilon_i, \varepsilon_i \sim \mathcal{N}(0, \sigma_n^2)$$

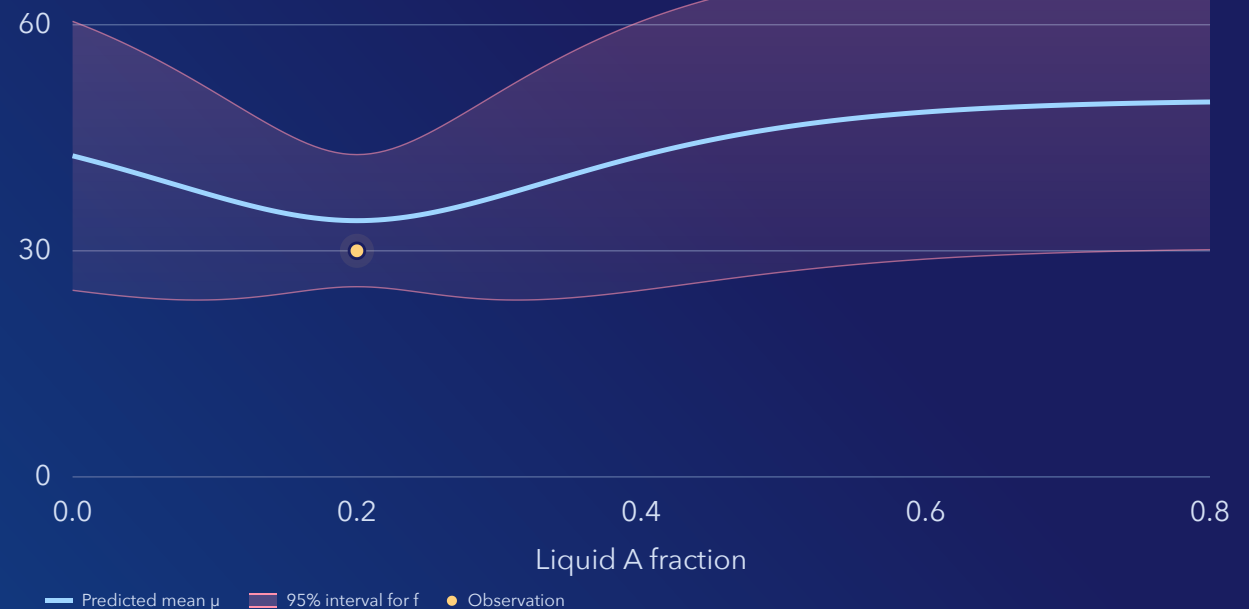
Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.

Model assumption: for teaching, variation in measured MSE is approximated by independent Gaussian noise.

The influence spreads to similar recipes through the kernel.

Prior becomes posterior

Color MSE ↓
90



1D mixture slice: $B = 0.20$; $C = 0.80 - A$. With noisy observations, the mean need not pass exactly through the measured point.

The kernel decides *who informs whom*.

Nearby recipes are more related in this example. The **length scale** ℓ controls how far an observation's influence reaches.

$$k\left(\underline{x}, \underline{x}'\right)=100\left(1+\underline{z}+\frac{\underline{z}^2}{3}\right) \exp \left(-\underline{z}\right)$$
$$\underline{z}=\frac{\sqrt{5}\left|\underline{x}-\underline{x}'\right|}{\ell}$$

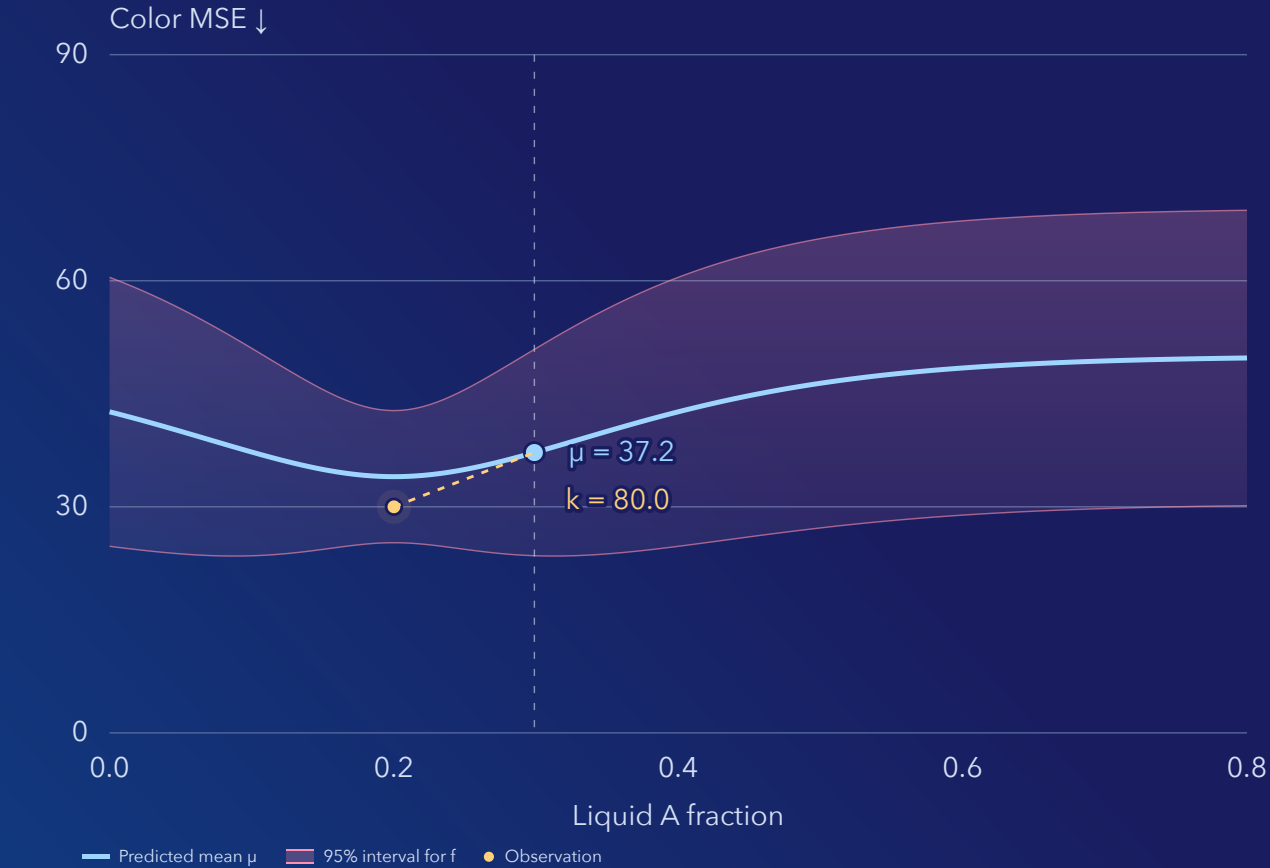
Matérn 5/2: the notebook's kernel family. Prior variance 100; initial length scale 0.18, adjustable below. The notebook fits its kernel parameters.

Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.

Length scale ℓ 0.18

Longer length scale: slower variation across recipes and wider influence.

Move the length scale



1D mixture slice: B = 0.20; C = 0.80 – A. The gold observation stays fixed. Only the model's similarity assumption changes.



GP 04 · KERNEL CHOICES

One GP framework. *Many kernels.*

A kernel encodes the patterns we expect. Matérn 5/2 allows less smooth variation than RBF. Here are two other choices.

$$d = |x - x'|$$

Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.

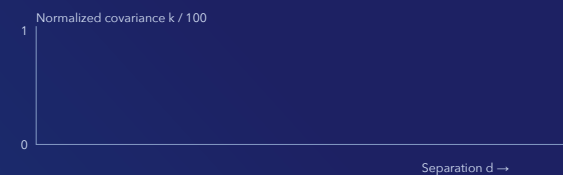
For this notebook, keep Matérn 5/2. A periodic kernel would require evidence of a repeating pattern.

Two other covariance patterns

RBF · extremely smooth

Similarity fades with distance. Useful when you expect very smooth changes.

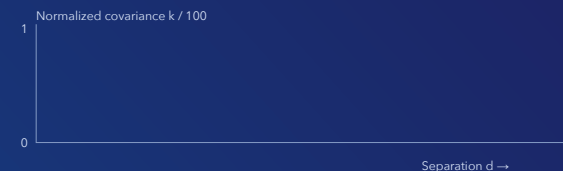
$$k(x, x') = 100 \exp\left(-\frac{d^2}{2\ell^2}\right)$$



Periodic · repeating patterns

Similarity returns every period p . Useful for cycles, such as a repeating time signal.

$$k(x, x') = 100 \exp\left(-\frac{2\sin^2\left(\frac{\pi d}{p}\right)}{\ell^2}\right)$$



Illustrative covariance curves, not objective values or measured data. Both formulas use amplitude 100; plots show normalized covariance.



GP 05 · CALCULATING M

Mean = prior + a data correction.

At a new recipe $A \approx 0.299$, the covariance with our observation is **80**. Noise reduces how strongly we trust that observation.

$$\mu_{**} = 50 + \frac{80}{100 + 25}(30 - 50) = 37.2$$

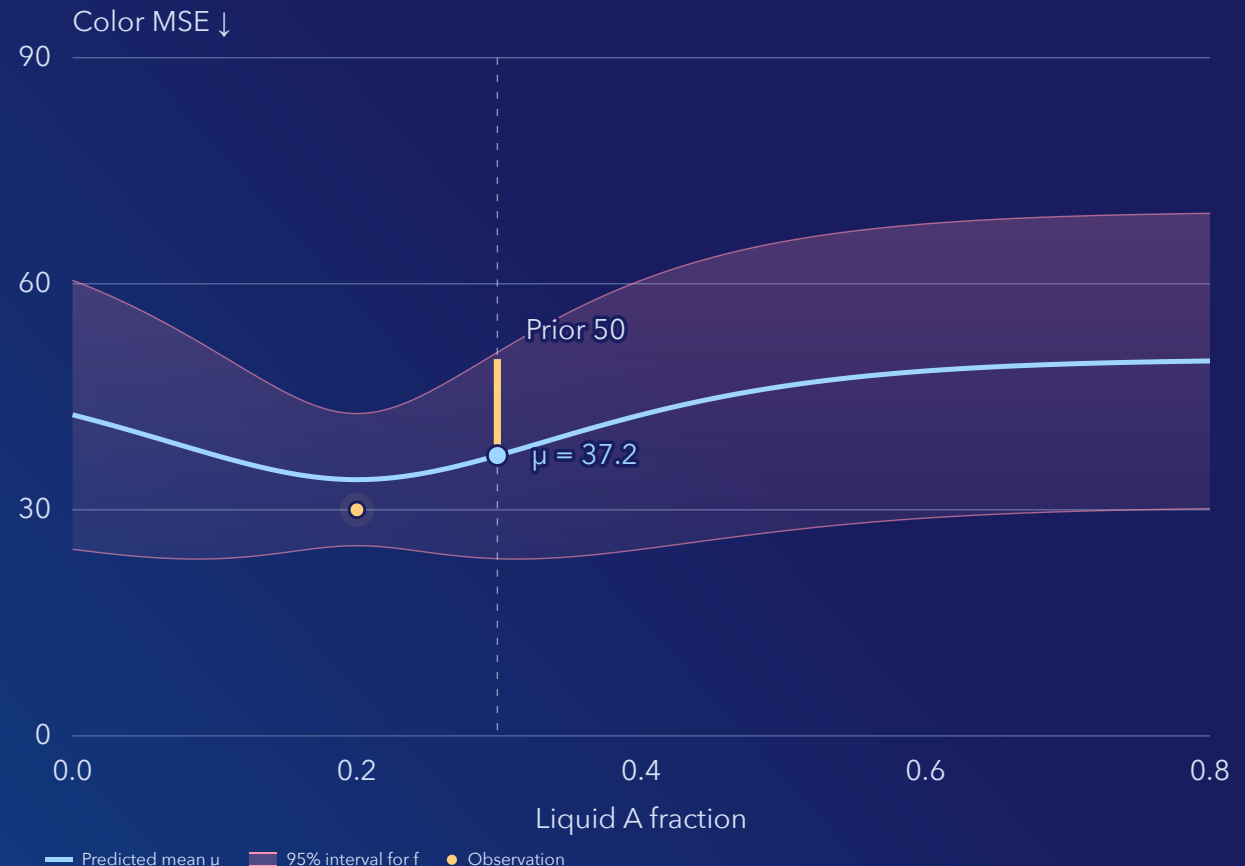
$$\mu(x_{**}) = m(x_{**}) + k_{**}^T C^{-1}(y - m)$$

Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.



The weight is $80 / 125 = 0.64$. It moves the prior 64% of the way toward 30.

A weighted correction at the new recipe



1D mixture slice: $B = 0.20$; $C = 0.80 - A$. $C = K + \sigma_n^2 I$. Here, one observation makes C the scalar $100 + 25$.

GP 06 · CALCULATING Σ

Uncertainty = *what the data leave unknown.*

Start with the prior variance. Subtract the part explained by the observations. Then take the square root to obtain σ .

$$\sigma_{**}^2 = 100 - \frac{80^2}{125} = 48.8$$

$$\sigma_{**} = \sqrt{48.8} \approx 6.99$$

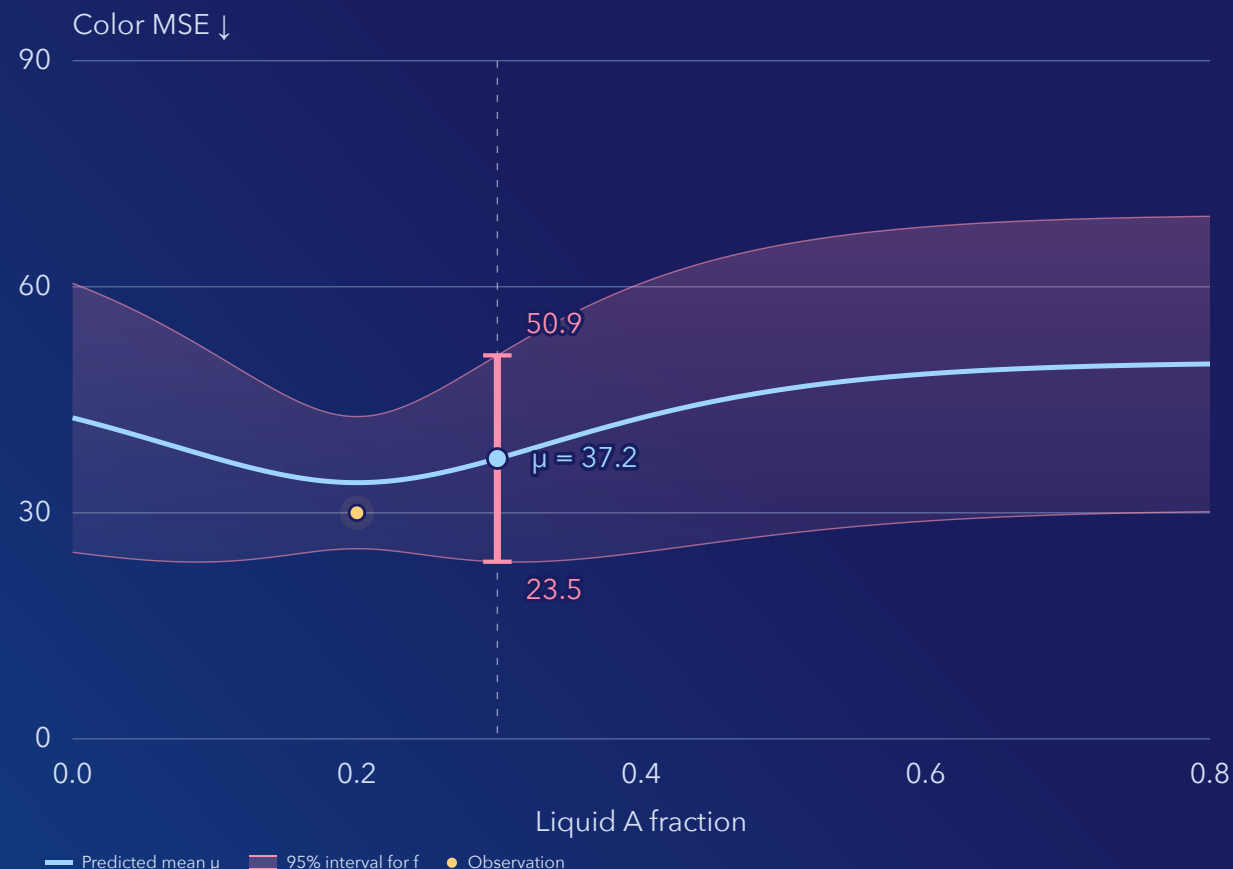
$$\sigma_{**}^2(x_*) = k(x_*, x_*) - k_{**}^T C_{**}^{-1} k_{**}$$

Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.



The 95% interval at this recipe is approximately $37.2 \pm 1.96 \times 6.99$.

The interval at one untested recipe



1D mixture slice: B = 0.20; C = 0.80 – A. For the underlying error $f(x^*)$: approximately [23.5, 50.9] MSE units.

GP 07 · MEASUREMENT NOISE

Noisier measurements give *weaker evidence*.

Increase the measurement noise. The model trusts the observation less: the mean moves back toward the prior and the interval widens.

$$\text{Var}(f_{**} | \mathcal{D}) = \sigma_{**}^2$$

$$\text{Var}(y_{**} | \mathcal{D}) = \sigma_{**}^2 + \sigma_n^2$$

Independent Gaussian measurement noise.

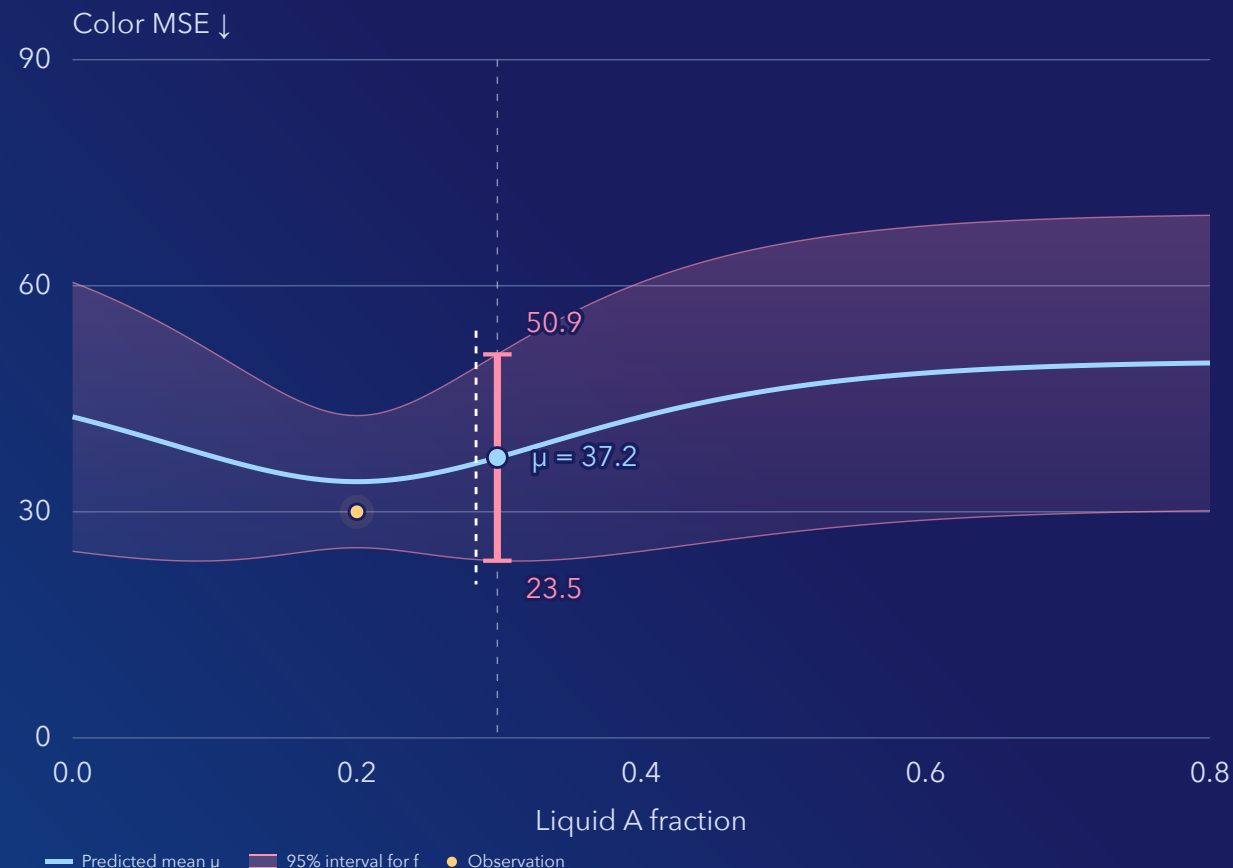
Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.

Noise standard deviation σ_n 5.0

At $A \approx 0.299$: $\mu = 37.2$ $\sigma = 6.99$

Uncertainty about the underlying response and measurement noise are different.

Same observation, different noise assumptions



1D mixture slice: $B = 0.20$; $C = 0.80 - A$. Solid interval: underlying response. Dashed interval: a future noisy measurement.

GP 08 · REPEATED EXPERIMENTS

Every new point updates the whole curve.

Add the measurements one at a time. Watch the mean move and the uncertainty shrink around the tested recipes.

Data → updated $\mu(\underline{x})$, $\sigma(\underline{x})$

Six illustrative experiments. The kernel stays fixed.

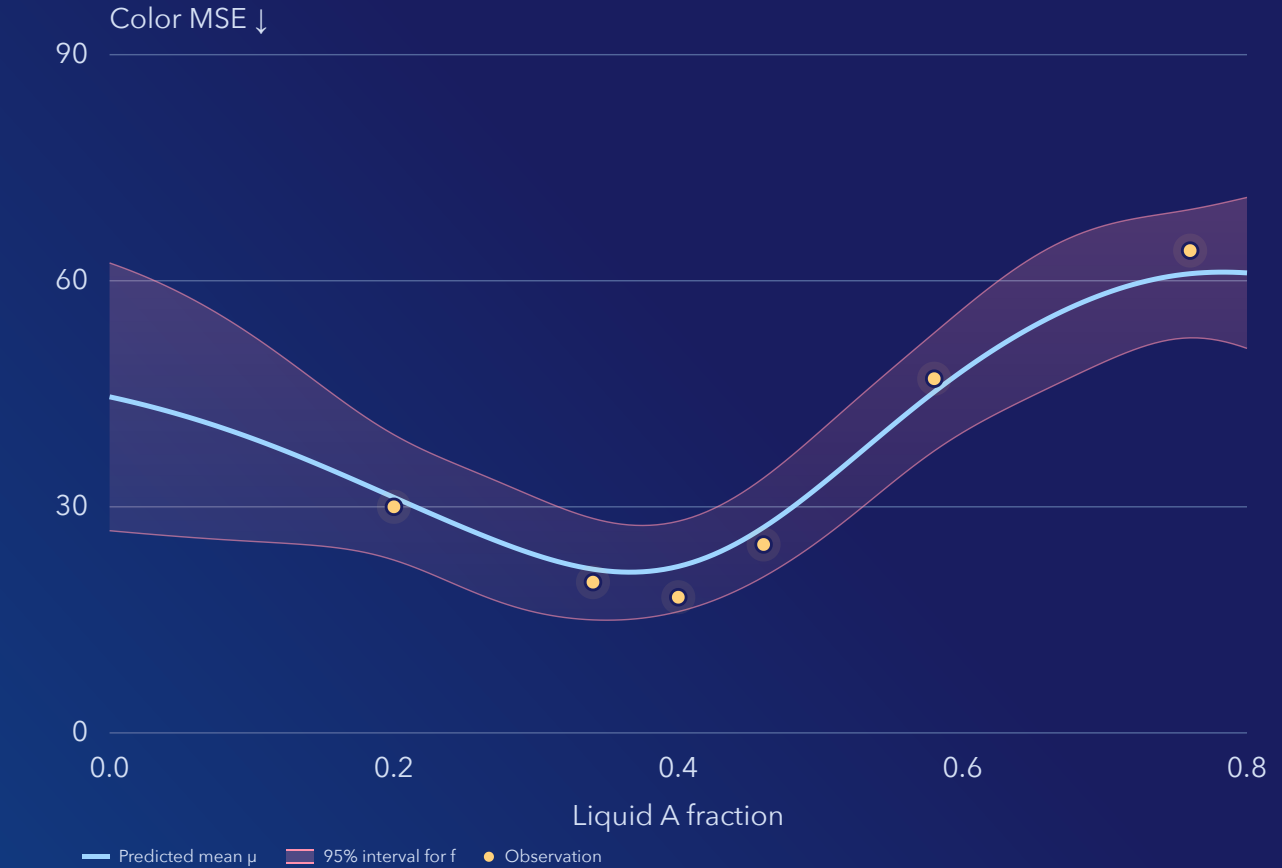
Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.



6 / 6 observations

More local evidence usually means less local uncertainty.

Learning over six experiments



1D mixture slice: $B = 0.20$; $C = 0.80 - A$. We refit the GP after each point. These six recipes are a fixed teaching sequence, not EI selections.

Acquisition Function: Choose where to *learn next*.

Expected Improvement combines the predicted error with uncertainty. It prefers a recipe with a promising chance of beating the best result.

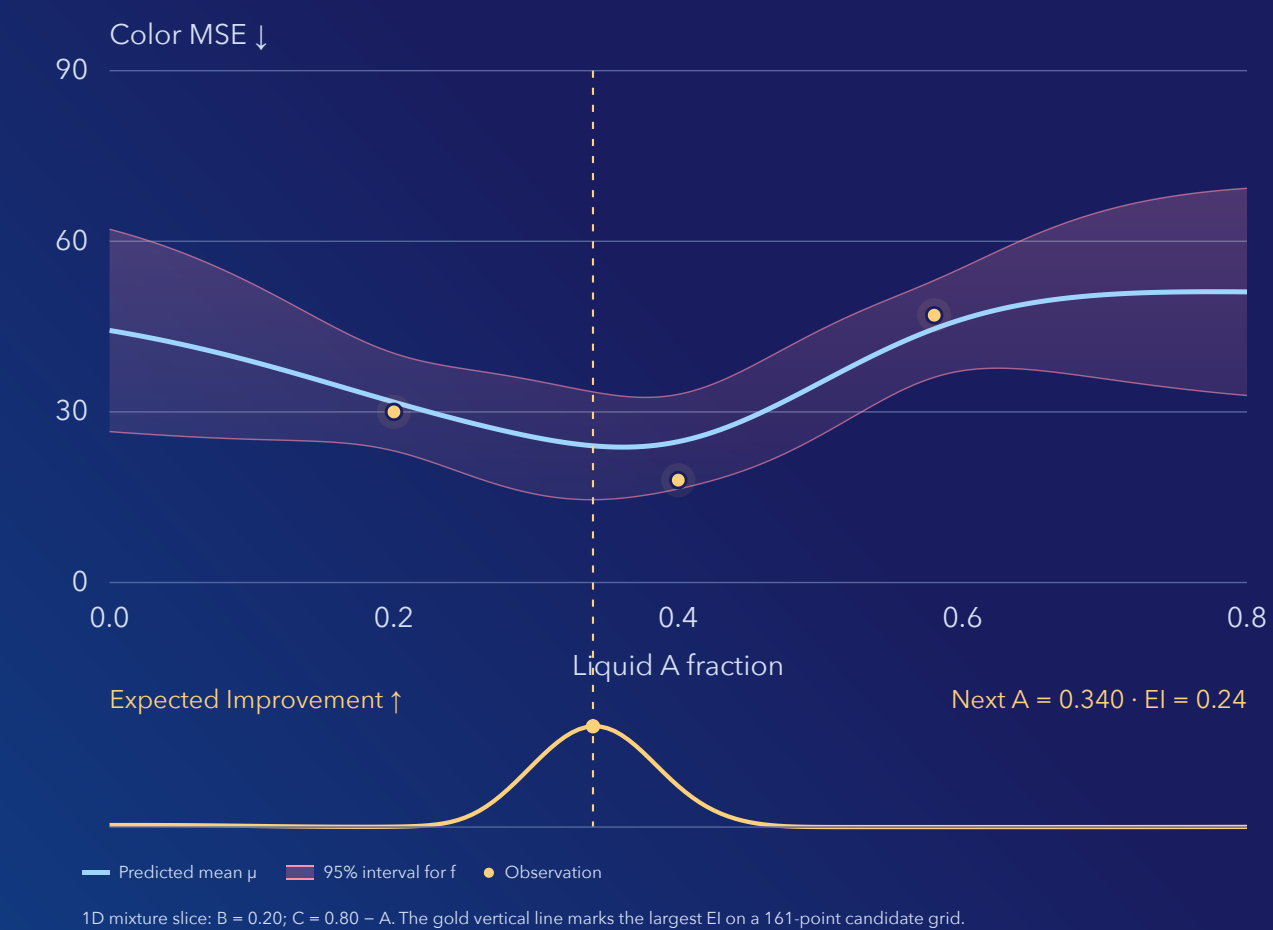
$$I(\underline{x}) = \max\{y_{\min} - f(\underline{x}) - \xi, 0\}$$
$$EI(\underline{x}) = \mathbb{E}[I(\underline{x}) \mid \mathcal{D}]$$

Minimization; $\xi = 0.01$. Larger EI is better.

Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.

The next recipe comes from the acquisition function, not just the lowest mean.

Prediction above, Expected Improvement below



GP 10 · TRY IT YOURSELF

You choose a point. *The model responds.*

Add an imagined measurement to the upper MSE graph. Watch the GP prediction and the EI curve below update from the same observations.

Click the graph, or enter a recipe and MSE.

All added values are local teaching data. No robot is connected.

Model limit: actual MSE cannot be negative, but an unconstrained GP can predict below zero. The graph clips predictions and intervals outside 0-90.

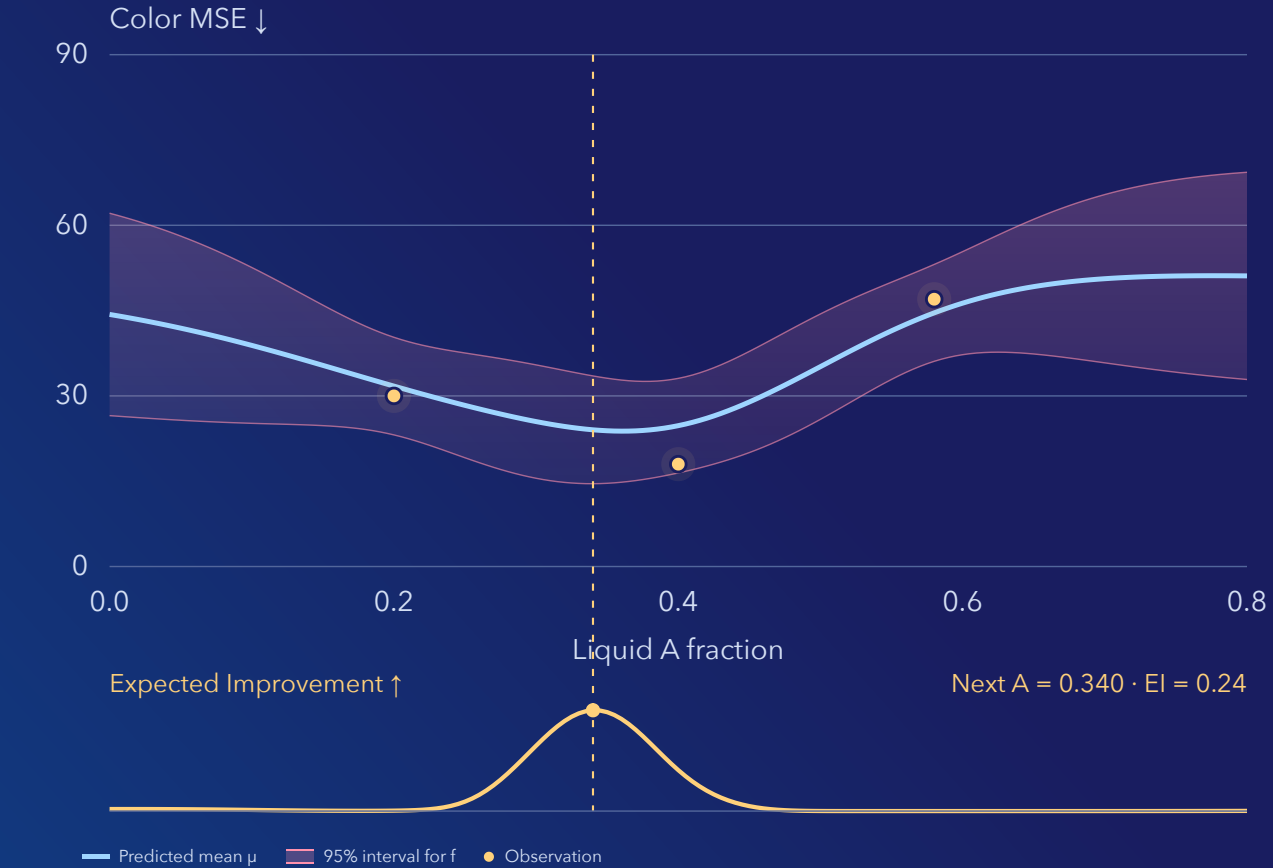
Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.

A MSE

3 / 12 observations

Try a surprising observation in a region where the model is already confident.

Your experiments update both the GP and Expected Improvement



1D mixture slice: B = 0.20; C = 0.80 – A. Gold marks the largest EI on the candidate grid ($\xi = 0.01$). The EI vertical scale adjusts automatically. Reset restores three observations.



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The computer chooses. *The robot executes.*

A CSV connects the proposed recipe to a destination well. The liquid-handling protocol must actually receive and use that recipe.

colorA · colorB · colorC · DispensePos

Responsibilities in the physical loop

Computer

- Proposes liquid volumes
- Writes the recipe CSV
- Sends protocol commands
- Processes the final image

OT-2

- Executes the protocol
- Dispenses into named wells
- Reports run status
- Provides the camera image

The attachment references color_mixing.py, which was not supplied.

A saved CSV alone does not dispense any liquid.



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A photograph becomes *a plate image*.

Capture after the protocol finishes. Correct the perspective and isolate the plate before extracting well colors.

Capture → correct perspective → crop

Corner coordinates and crop boundaries depend on the camera setup.

The notebook's image-processing pipeline

01 · Original camera view



02 · Perspective corrected



Historical notebook outputs. Consistent illumination matters.

Each well becomes *an RGB* *measurement.*

Divide the cropped plate into an 8 × 12 grid. Select pixels inside each well and take the median in each color channel.

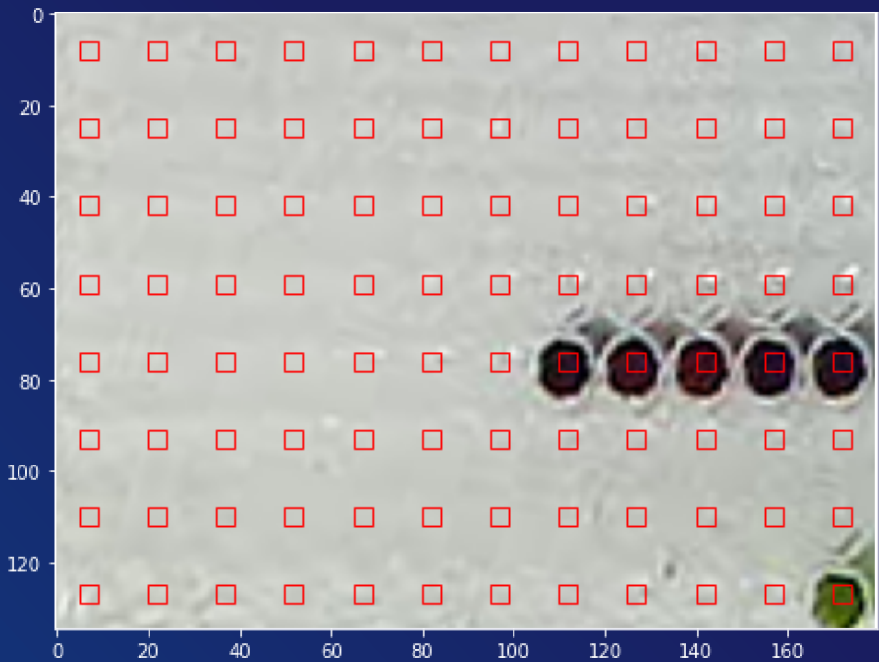
Code: D1–D5 are samples
H12 is the target reference

Initial sampling: Start with 5 random recipes (D1–D5), each totaling approximately 300 μ L after rounding. Their measured MSE values form the initial dataset before BO selects the next recipe.

Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.

Check the physical well mapping before trusting the extracted numbers.

Regions of interest inside the wells



ROI = region of interest. The saved image's filled wells differ from the current D1–D5 selection.



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One more experiment. *One more data row.*

After obtaining a new RGB measurement, calculate its MSE and append the observation. Refit the GP with all the data.

X: n × 3 RGB_exp: n × 3

Y: n × 1

Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.

Keep the recipe, RGB and error in exactly the same row order.

The data used in each iteration

01 `xnext(X, Y)` proposes volumes

02 Divide by 300 to store fractions

03 Obtain RGB and calculate the MSE

04 Append matching rows to X, RGB_exp and Y

05 Fit again using the enlarged dataset

`scikit-optimize: base_estimator="GP", acq_func="EI".`



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A worse test can *still teach the model.*

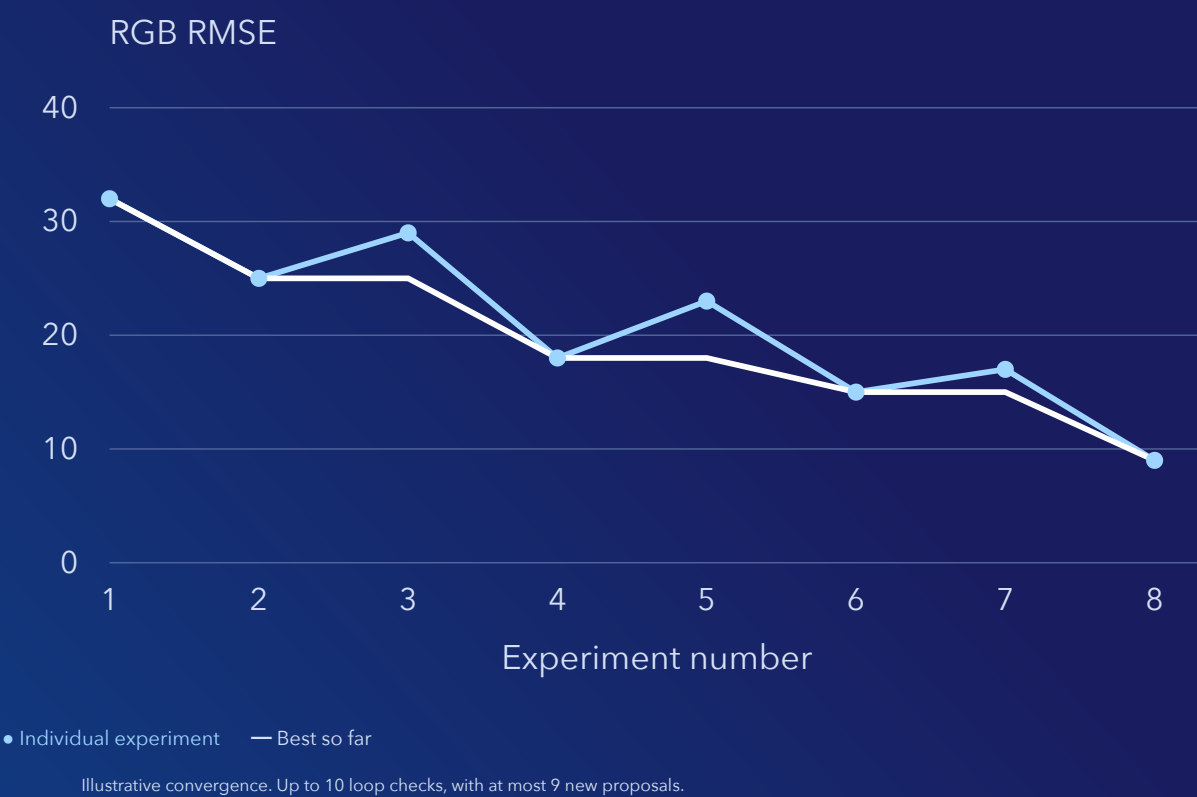
Individual experiments may have larger errors. The best observed error can only stay the same or decrease.

Success: the lowest-MSE recipe
is within ± 8 in every RGB channel

Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.

RMSE below 8 alone does not guarantee that each channel passes.

Individual results and best so far



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The notebook includes *a simulated continuation.*

Initial values come from image analysis. Later rounds use `RGB_func()` to generate toy RGB measurements.

A full robot loop must replace
`RGB_func()`.

Do not interpret mixed physical and simulated data as a validated lab optimization.

Where the demo changes from physical to simulated

Initial measurements

Plate image
Well regions
Median RGB
Error against H12

Later rounds

Next recipe CSV
`RGB_func()` simulation
Simulated error
Dataset update

The active target is the RGB extracted from H12.

μ predicts. σ tells us *what is still uncertain*.

The kernel links recipes. Observations update the prediction. Bayesian optimization uses both μ and σ to choose its next experiment.

Prediction + **uncertainty**
guide the next experiment

With one observation and a fixed GP prior, which change reduces the observation's influence on the posterior mean?

Keep the tested recipe and its measured MSE unchanged. Change only the assumed noise variance.

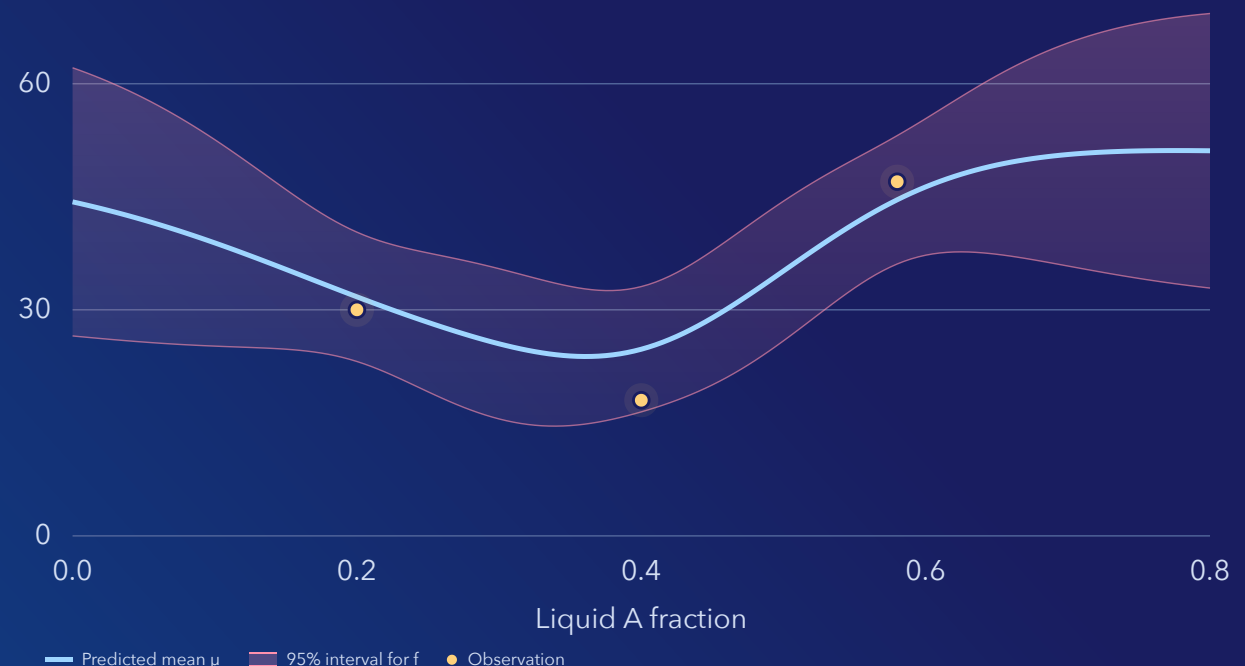
A. Increase the measurement-noise variance

B. Decrease the measurement-noise variance

The noise assumption controls how strongly a measurement updates the prior.

Ready for the code walkthrough

Color MSE ↓
90



1D mixture slice: B = 0.20; C = 0.80 – A. Notebook landmarks: `optimizer_func()`, `xnext()`, `MSE_func()`, then the measurement update.

Now open *the notebook.*

Follow the same experiment through setup, decision, execution, measurement and update.

What is x? What is Y?
Who chooses the next experiment?

Hover or tap an underlined symbol to explain it. Keyboard: Tab, then Enter.

Answers: recipe fractions, MSE values, and the acquisition rule based on the GP.

Your code walkthrough map

- 01 TOTAL_VOLUME, X_init
- 02 optimizer_func(), xnext()
- 03 upload_protocol(), execute_protocol()
- 04 image_slicing(), median RGB
- 05 MSE_func(), X, Y, RGB_exp

The original notebook remains unchanged.